MIL-A-45130A(Ord)
30 December 1959
SUPERSEDING
MIL-A-45130 (Ord)
26 June 1958

MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL, AM-1057/MS

1. SCOPE

1.1 This specification covers one type of electronic control amplifier, hereinafter referred to as "amplifier". The amplifier is a component of the antenna pedestal used in the antiaircraft fire control and guided missile systems listed as outlined in 6.1.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the manufacture and inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1285

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isfactory ... imperfect workmanship. (Operation within the specified limits of sal

DRAWINGS

Ordnance Corps

E7620653	Amplifier, Electronic Control, AM-1057/MS
F7614590	Amplifier, Electronic Control, AM-1057/MS, Schematic Diagram

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Production model. - At the beginning of production, one amplifier representative of production, shall be made available to the Government to determine if the amplifier meets all requirements of this specification.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction. - The amplifier shall be manufactured in accordance with Drawing E7620653, "Amplifier, Electronic Control, AM-1057/MS", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential. - The design and construction of the amplifier shall insure that all external parts are ground potential at all times.

3.4 General specification. - The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron tubes
(f) Workmanship

3.5 Environmental.

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3.5.1 Vibration. - The amplifier shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second, return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration, the amplifier shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperatures. - The amplifier shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.5.3 Storage temperatures. - The amplifier shall show no evidence of physical failure and shall meet all performance requirements of this specification at the standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of 160 degrees and minus 80 degrees Fahrenheit.

3.6 Electrical.

3.6.1 Continuity. - The amplifier shall demonstrate continuity in accordance with the electrical schematic, Drawing F7614590.

3.6.2 Dielectric strength. - There shall be no puncture of insulation, no sparkover or decrease in dielectric strength when the insulated areas of the amplifier are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than 5 seconds.

3.6.3 Insulation resistance. - Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.7 Performance. - When set up in accordance with pretest information (section 4), the amplifier shall meet the performance as specified herein. All symbols referred to in the following paragraphs are those shown on Drawing F7614590.

3.7.1 Amplifier stability and saturation.

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3.7.1.1 When power is applied, but no signal voltages, and an oscilloscope connected to terminals 5 and 9, in turn, the amplifier shall exhibit sweep lines of no amplitude.

3.7.1.2 When a 33-volt, 4000 cps signal is applied, there shall be no flattening at the top of the applied sine wave of the amplifier output or indication of oscillation at the peak of the wave on either channel.

3.7.2 Amplifier gain. - The voltage gain on each channel shall be between 1.00 and 1.10.

3.7.3 Comparison of channels. - The output voltages at terminals 5 and 9 shall not differ by more than 0.18 volt.

3.7.4 Interchannel crosstalk.

3.7.4.1 The output voltage at terminals 5 and 7 shall not be greater than 0.1 volt.

3.7.4.2 The output voltage at terminals 9 and 11 shall not be greater than 0.1 volt.

3.8 Identification marking. - Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

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(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Classification of tests . - The inspection and testing of the amplifier shall be classified as follows:

(a) Production model test
(b) Acceptance test

4.3 Production model tests.

4.3.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

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4.3.2 The test shall be conducted at the contractor's facility or Government approved laboratory, using test equipment designed or procured for use on the remaining items.

4.3.3 The model shall meet all the requirements of this specification using the inspection method prescribed herein.

4.3.4 Disposition of rejected production model. - Should the production model fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the production process. When corrections in the process have been made, a new production model shall be completely inspected by the contractor and witnessed by the Government inspector. The rejected production model shall be corrected, where applicable, and shall become part of the first production lot.

4.4 Inspection provisions.

4.4.1 Inspection of components. - All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 rms or better, threads, identification marking, and all dimensions on which the tolerances indicate a critical requirement. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.4.1.1 Sampling inspection. - Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with table 1.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0-25	Use 100% Inspection		
26-35	20	0	1
36-50	25	0	1
51-150	30	0	1
151-300	30	1	2
301-over	60	2	3

6.

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4.4.1.2 Disposition of rejected lots. - Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection of the characteristics that caused the lot to be rejected.

4.4.2 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.4.3 Lot formation. - Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.4 Acceptance tests.

4.4.4.1 Classification of defects. - Classification of defects in tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.4.4.2 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection use level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105
AQL value is expressed in percent defective

MAJOR: AQL 4.0 %	Req't Par.	QA Par.	Test Method
1. Vibration	3.5.1	4.6.1	Standard Measuring Equipment (SME)

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the Engineering Division, San Antonio ALC/MMEDQ, Kelly AFB, Texas 78241 by

TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 4.0%		Req't Par.	QA Par.	Test Method
2.	Physical failure, post vibration	3.5.1	4.6.1	Visual-Tactile
3.	Operating temperatures	3.5.2	4.6.2	SME
4.	Storage temperatures	3.5.3	4.6.3	SME
5.	Physical failure, post storage temperature	3.5.3	4.6.3	Visual-Tactile

TABLE III - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL value is expressed in percent defective

MAJOR: AQL 1.5%		Req't Par.	QA Par.	Test Method
1.	Completeness of assembly	Per dwg.	4.8	Visual
2.	Continuity	3.6.1	4.6.4.1	SME
3.	Dielectric strength	3.6.2	4.6.4.2	SME
4.	Insulation resistance	3.6.3	4.6.4.3	SME
5.	Amplifier stability and saturation	3.7.1	4.7.4	SME
6.	Amplifier gain	3.7.2	4.7.5	SME
7.	Comparison of channels	3.7.3	4.7.6	SME
8.	Interchannel crosstalk	3.7.4	4.7.7	SME
9.	Identification marking	3.8	4.9	Visual

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4.5 Inspection equipment.

4.5.1 Contractor provisions. - Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.5.2 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.5.2.1 In addition to the above accuracy requirements, the test equipment and the related testing components prescribed below shall conform to the following:

Item		Requirement
Oscillator		Variable voltage output 0-40 volts, 4000 cps.
	Accuracy:	Frequency, within plus or minus 5 percent. Output voltage, within plus or minus 0.001 volts. Output impedance, within plus or minus 5 percent unit under test.
Vacuum tube Voltmeter (VTVM)	Accuracy:	Voltage, within plus or minus 2 percent over the entire range. Input impedance, 0.5 megohms plus or minus 1 percent shunted by 30 micro-microfarad plus or minus 1 percent.

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Item (Cont'd)		Requirement
Oscilloscope	Input impedance:	300,000 ohms parallel by 30 micro-microfarads. With probe, 3 megohms parallel by 12 micromicrofarads.
	Output:	Trigger pulse plus 25 volts, 4 micro-second duration, occuring 300,800 or 2,000 times per second, rising from 10 to 90 percent of full amplitude within 1/2 microsecond, Sawtooth voltage plus 150, available through 100,000 ohms in series with 50,000 micromicrofarads.
	Frequency range:	1 cycle to 3 megacycles per second. Response drops 0.3 at each end-point.
	Sensitivity:	A signal of 0.1 volt to produce a standard deflection with maximum ampli-fication.
	Sweep time:	Continuously adjustable from 0.5 to 50,000 microseconds per inch. Start-Stop, each sweep started by signal pulse independent of preceding. Any portion of sweeps nominally over 10 microseconds may be delayed and ex-panded about ten times for detailed ex-amination of signal.
	Rise time:	0.08 microseconds.
	Probe:	Input impedance may be increased by ten times at expense of a ten to one re-duction in sensitivity by use of the probe. Use of probe also increases the voltage range to 450.
Resistor		5000 ohms plus or minus 10 percent, 1/2 watt.
Power supply		plus 250 volts, plus or minus 5 volts, 0.10 ampere.
Power supply		120 volts plus or minus 5 volts, 400 cycle, single phase, 1 ampere.

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4.6 Test methods and procedures.

4.6.1 Vibration. - The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.5.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified, with the amplifier rigidly mounted in its normal operating position. Subsequent to the vibration test, the amplifier shall show no damage or defect resulting from the test by a visual and tactile examination. The amplifier than shall be inspected as specified herein to determine conformance with the requirements specified.

4.6.2 Operating temperatures. - The test equipment used in conducting the temperature tests shall be such that the units under test shall not exceed fifty percent of the internal volume of the test chamber, and no portion shall be closer than six inches from any surface of the chamber.

4.6.2.1 The units being tested shall be installed in a test chamber in a manner similar to its normal operating position. The amplifier shall be exposed to the conditions specified in 3.5.2 utilizing a test chamber capable of maintaining the specified temperatures until the unit under test is thermally stabilized. When thermal stabilization of the unit is reached at the specified conditions, respectively, the united under test shall be inspected in accordance with the prescribed methods and procedures to determine conformance with the requirements specified herein.

4.6.3 Storage temperatures. - The amplifier to be tested shall be installed in a test chamber in a manner similar to its normal operating position. The test shall then be conducted in accordance with Procedure II of Specification MIL-F-13926, to the conditions specified in 3.5.3. Upon completion of the specified exposure, the unit under test shall show no evidence of damage by a visual and tactile examination. The unit shall then be inspected in accordance with the prescribed methods and procedures to determine conformance with the requirements specified herein.

4.6.4 Electrical.

4.6.4.1 Continuity. - A continuity check of the amplifier circuits shall be made using standard measuring equipment to determine conformance with the requirement specified in 3.6.1.

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4.6.4.2 Dielectric strength.- The amplifier shall be tested in accordance with method 301 of Standard MIL-STD-202 to determine conformance with the requirements specified in 3.6.2.

4.6.4.3 Insulation resistance.- Immediately after satisfactorily withstanding the test for dielectric strength, the amplifier shall be given the insulation test in accordance with method 302, test condition B, of Standard MIL-STD-202 to determine conformance with the requirements specified in 3.6.3.

4.7 Performance.- The amplifier, to determine compliance with 3.7, shall be set up as outlined herein.

4.7.1 Pretest information.- Terminals 1 and 2 of connector P1 of the amplifier shall be strapped and connected to the high side of an oscillator, with the ground side of the oscillator connected to terminals 3, 4 and 13 of P1.

4.7.2 Two dummy loads shall be constructed. Each load shall consist of a 1000 ohm plus or minus 5 percent, 10 watt resistor, a 0.3 microfarad condenser plus or minus 20 percent, 200 working volts dc, and a 0.02 henry, 4.1 ohm coil; all in parallel. One dummy load shall be connected between terminals 9 and 11 of P1, the other dummy load shall be connected between terminals 5 and 7 of P1.

4.7.3 A plus 250 volt dc power supply shall be connected to terminals 8 of P1 and terminals 7, 11, and 13 shall be grounded. A 120 volt power supply shall be connected between terminals 14 and 15 of P1.

4.7.4 Amplifier stability and saturation.

4.7.4.1 An oscilloscope shall be connected, in turn, to terminals 5 and 9 of P1 with the input terminated. Only a sweep line with no amplitude shall appear on the test ocsilloscope.

4.7.4.2 With 33 volts, 4000 cps signal applied from the test oscillator to the input of the amplifier under test, there shall be no flattening of the top of the applied sine wave at the amplifier output or indication of oscillation at the peak of the sinusoid on either channel.

4.7.5 Amplifier gain.- The voltage gain on each channel shall be between theprescribed value of 3.7.2 as measured with a VTVM.

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4.7.6 Comparison of channels. - With 10 volts at the inputs of the amplifier under test, the output voltages at terminals 5 and 9 of P1 as measured with the VTVM shall not differ by more than the value specified in 3.7.3.

4.7.7 Interchannel crosstalk.

4.7.7.1 The strap between terminals 1 and 2 of P1 shall be disconnected. A 5000 ohm plus or minus 10 percent 1/2 watt resistor shall be connected between terminals 1 and 3 of P1. The test oscillator shall be connected between terminals 2 and 4 of P1 and 30 volts; 4000 cycles shall be applied, at the input. The output voltage measured between terminals 5 and 7 shall not be greater than the value specified in 3.7.4.1.

4.7.7.2 Procedure specified in 4.7.7.1 shall be repeated for the other channel by connecting the 5000 ohm resistor between terminals 2 and 4 of P1. The oscillator shall be connected to terminals 1 and 3 of P1, and 30 volts applied. The output voltage measured between terminals 9 and 11 shall not be greater than the value specified in 3.7.4.2.

4.8 Check for completeness. - In addition to the performance and accuracy tests above, the finished unit shall be checked against the assembled view drawings to ascertain that all visible parts shown on the drawing and listed on the list of parts have been assembled and that all locking and fastening screws, pins, nuts, etc., are in place and secured as required by the drawings and specifications.

4.9 Identification marking. - Marking for identification shall be visually inspected for conformance in accordance with Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. - The amplifier shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 P109. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The amplifier is intended to be used as a low-impedance matching device between a signal source from the acquisition azimuth resolver and a signal output that shall be transmitted to subsequent resolvers and plan positioning indicators (PPIs'). It is also intended to furnish more power than a resolver alone is capable of supplying.

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6.1.1 Application. - The amplifier is a component of the assemblies and systems listed below:

Assemblies	Systems
Antenna - Receiver - Transmitter Group, Acquisition OA-653/MS	Antiaircraft Fire Control System M33C(90MM) and M33 (120MM)
Antenna - Receiver - Transmitter Group, Acquisition OA-653/MS	Radar Course Directing Control AN/MSW-1 and AN/MSW-3
Antenna - Receiver - Transmitter Group, Acquisition	Antiaircraft Guided Missile System Nike Hercules
Antenna - Receiver - Transmitter Group, Acquisition OA-1596/T	Guided Missile Fire Control System AN/MSG-3A

6.2 Ordering data . - Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (See 3.1)

6.3 Definitions.

6.3.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.2 Tactile. - Skillful manipulation or sensing.

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